Ramin Akbari

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EDUCATION

Metropolitan State University of Denver Expected May 2027

B.S. in Computer Science GPA: 3.28

Relevant Coursework: Data Structures & Algorithms, Software Development Methods, Networking & Distributed Computing, Mobile App Development, Computer Vision, Computer Organization

TECHNICAL SKILLS

Languages (Proficient): Java, C++

Languages (Familiar): JavaScript, HTML/CSS, Python, Kotlin, ARM Assembly

Frameworks & Libraries: WebExtensions API, Android SDK, Jest, OpenCV, NumPy

Tools: Git/GitHub, GitHub Actions, Linux (Ubuntu), Bash, ESLint, Android Studio, VS Code, Eclipse

Concepts: Data Structures, OOP, Socket Programming, Unit Testing, CI/CD, Agile/Scrum

PROJECTS

Anonymous Theme Switcher | *JavaScript, Jest, GitHub Actions* | Firefox Add-ons | GitHub Spring 2026

- Built and published a Firefox extension with a 5-person Agile team, backed by 45 Jest tests and CI.
- Developed theme detection and grouping with the WebExtensions management API, sorting installed themes into collapsible groups saved across sessions.
- Implemented live hover previews that apply a theme instantly and restore the original on mouse-out, with click-to-lock logic to keep the user's choice.

Digital Signage Display | *JavaScript, HTML/CSS* | GitHub Spring 2026

- Built a config-driven signage web app with a 5-person team, combining 6+ public APIs into one display.
- Fixed CORS errors blocking the news feeds by routing RSS requests through the rss2json API after a standard CORS proxy failed.
- Developed the weather widget, showing current conditions and a 3-day forecast from Open-Meteo.

Lane Detection | *Python, OpenCV, NumPy* | GitHub Spring 2026

- Built a classical computer vision pipeline (no ML) that detects lane markings in KITTI road images.
- Combined Sobel edges, thresholding, masking, and morphology; tuned kernels to keep thin lane lines.
- Fit lane lines with the Hough transform and tested on clear and partially blocked road scenes.

Car Vending Machine | *Java* | GitHub Spring 2025

- Built a Java inventory system for a multi-floor car vending machine with search, sales, and a wash queue.
- Refactored storage from a 2D array to a location-keyed HashMap for O(1) lookups, inserts, and removals.
- Replaced duplicated sort methods with one Comparator-based sort and added a class hierarchy.